Lecture 23

Thursday, 6 April 2023 1:32 PM

RDE: $-a\bar{z}^2\frac{dc}{d\bar{z}} = \frac{d^2c}{d\bar{z}^2}$, $\bar{z} = \left(\frac{\Omega^{1/2}}{\nu^{1/6}D^{1/3}}\right)z$

BC's: @ $\bar{z}=0$, $c=0$; $g=\frac{dc}{d\bar{z}}$

@ $\bar{z}=\infty$, $c=c_\infty$

$-a\bar{z}^2 g = \frac{dg}{d\bar{z}} \Rightarrow -a\bar{z}^2 d\bar{z} = \frac{1}{g}dg$

$-\frac{a\bar{z}^3}{3} = \ln g$

$\frac{dc}{d\bar{z}} = g = C_1 \exp\left(\frac{-a\bar{z}^3}{3}\right)$ → Use gamma functions to solve for C

↓ Add in the BC's

$c_\infty = C_1\left(\frac{\Gamma(\frac{1}{3})}{3^{1/3}a^{1/3}}\right) \Rightarrow C_1 = \frac{c_\infty 3^{2/3}a^{1/3}}{\Gamma(\frac{1}{3})}$

$\frac{dc}{dz} = \frac{c_\infty 3^{2/3}a^{1/3}}{\Gamma(\frac{1}{3})}\exp\left[\frac{-a}{3}\bar{z}^3\right]$

@ $\bar{z}=0$ $\left.\frac{dc}{dz}\right|_{\bar{z}=0} = 0.62\, c_\infty$

$$\boxed{\left.\frac{dc}{dz}\right|_{z=0} = \left(\frac{\Omega^{1/2}}{\nu^{1/6}D^{1/2}}\right)(0.62)\,c_\infty}$$

↓

From Ch 5: $N = -D\left.\frac{dc}{dz}\right|_{z=0}$

$N = -D c_\infty\left(\frac{0.62\,\Omega^{1/2}}{D^{1/3}\nu^{1/6}}\right)$

Normally, Flux @ $i_{lim} \Rightarrow N = -D\frac{c_\infty}{\delta}$

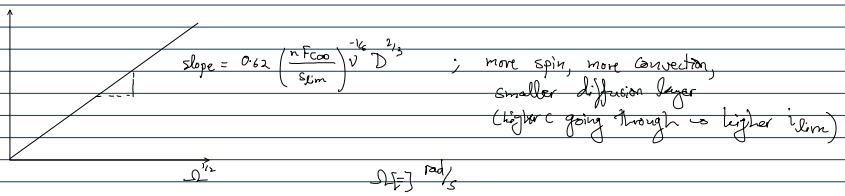


$\therefore$ Similarity $\Rightarrow \delta = \left(\frac{0.62\,\Omega^{1/2}}{D^{1/3}\nu^{1/6}}\right)^{-1}$

From Faraday's Law:

$$\boxed{\frac{\delta_{lim}}{nF}i_{lim} = \frac{Dc_\infty}{\delta} = 0.62\,D^{2/3}\nu^{-1/6}\Omega^{1/2}c_\infty}$$ Levich Equation!

If we can accurately vary Ω (rotation rate), we can understand species diffusion



; more spin, more convection, smaller diffusion layer (higher c going through ↔ higher i_{lim})

Note: we can also define "Levich-type" eq. for non-lim current, as long as current dist. is uniform ($W_a \rightarrow$ large)

(You will just have to change the bounds of integration)

Then, $\frac{\delta i_{ang}}{nF} = \frac{D(c_b - c_s)}{\delta} = 0.62\,D^{2/3}\nu^{-1/6}\Omega^{1/2}(c_\infty - c_s)$

Often, we have charge transfer resistance

↳ This means current has contributions from MT + kinetics

↳ To separate out kinetics + MT effects on current, use "Koutecky-Levich" analysis

↳ To understand kinetic effects, we generally define current as function of potential

$i = nF c_s K(V)$

surface conc. ↑ ↳ potential dependent rate constant

$K(V) = K_0\exp(f(\eta))$

We need to get an expression for c_s in terms of i_{lim}:

$\frac{\delta i_{ang}}{nF} = \frac{D(c_\infty - c_s)}{\delta} = \frac{Dc_\infty}{\delta} - \frac{Dc_s}{\delta}$

$= i_{lim} - Dc_s$;

$$\frac{S\, i_{ang}}{nF} = \frac{D(C_\infty - C_0)}{\delta} = \frac{DC_\infty}{\delta} - \frac{DC_0}{\delta}$$

$$= i_{lim} - \frac{DC_0}{\delta}$$

$$i = i_{lim}\left(\frac{C_\infty - C_0}{C_\infty}\right)$$

$$C_0 = C_\infty - \frac{i}{i_{lim}} C_\infty = C_\infty\left(1 - \frac{i}{i_{lim}}\right)$$

$$\boxed{i = nF_6\, K(V)\, C_\infty \left(1 - \frac{i}{i_{lim}}\right)}$$

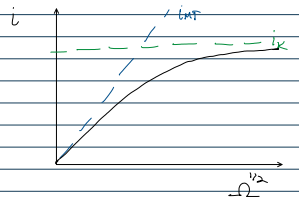
We can call this i_K → kinetic contr. to current

$i_{lim} = i_{MT}$ → mass transport contr. to current

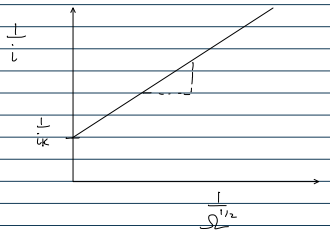
"Summing up resistors"

$$i = i_K\left(1 - \frac{i}{i_{MT}}\right) \Rightarrow i = \frac{i_K\, i_{MT}}{i_{MT} + i_K} \Rightarrow \boxed{\frac{1}{i} = \frac{1}{i_K} + \frac{1}{i_{MT}}}$$

↳ Koutecky-Levich Equation



If we tried to do Levich analysis, D might not be accurate → We can plot $\frac{1}{i}$ vs. $\Omega^{-1/2}$ (Linearize the system)



Primary Current Distribution for Rotating Disks

RDEs typically used near i_{lim}, but not always (w/ cases not near i_{lim})

↳ Does rotation rate of RDE affect primary current analysis?

No, because $\nabla C_j = 0$ for primary (conver. + diffusion don't matter, we just solve $\nabla^2\phi$)

+ Only difference is geometry, (solved by Neumann analytically):

$$\frac{i(r)}{i_{ang}} = \frac{0.5}{\sqrt{1 - (r/r_0)^2}} \qquad @\ z=0 \quad r > r_0 \Rightarrow \frac{\phi}{V} = 1 - \frac{2}{\pi}\tan^{-1}\sqrt{\left(\frac{r}{r_0}\right)^2 - 1}$$

+ Potential varies rapidly outside of disk close to disk

That is why to have

The potential drop in solution will affect CE & RE?
↳ How?

Position of CE is determined by this sort of analysis

If you are 3 disk radii away you have already gone through 90% of

Rotating Ring Disk Electrode (RRDE):

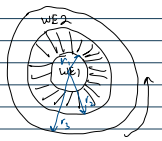
↳ We can add a second WE co-planar w/ disk to give added functionality to well-defined hydrodynamics of RDE

a measured potential drop

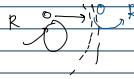
you have
a compensation!

↓

Luggin Cappillary



Most common application: "generator-collector" mode



Can get exactly how much / partial current going towards the desired reaction